BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 146 · Truth Tables and Combined Logic

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. How can you recognise a NOT gate symbol?

Q2. What does an encryption utility produce?
A. A smaller file
B. Ciphertext
C. A sorted folder
D. Extra storage space

Q3. “Publicly posting all personal records always improves privacy.” Correct this. Explain why.

Q4. A login must get details, check them and choose the next step. “Algorithmic thinking means guessing steps in a random order.” Correct this. Why does order matter?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Combined logic — Work out intermediate gate outputs before calculating the final result.

Operator precedence — Use brackets to clarify order when translating logic diagrams.

Three inputs — A truth table has $2^3 = 8$ unique input combinations.

Intermediate columns — Recording a gate's result reduces mistakes in multi-stage circuits.

New learning

Combined logic: Work out intermediate gate outputs before calculating the final result.

Operator precedence: Use brackets to clarify order when translating logic diagrams.

Three inputs: A truth table has $2^3 = 8$ unique input combinations.

Intermediate columns: Recording a gate's result reduces mistakes in multi-stage circuits.

Worked example

Complete Q = (A AND B) OR C for all eight input combinations.

1. List 000,001,010,011,100,101,110,111 in ABC order.

2. Find X=A AND B: 0,0,0,0,0,0,1,1.

3. Then Q=X OR C: 0,1,0,1,0,1,1,1.

4. Check rows with C=1 must have Q=1.

We Do · Practise with guidance

Guided 1

Find Q=(A OR B) AND C when A=1, B=0, C=0 and then C=1.

Worked method: Explain why AND is applied last.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Complete the condition so A with B, or C alone, raises an alert.

```
A = true
B = false
C = true
if ... then
    print("Investigate")
endif
```

write

Question 1

How many rows are needed for a 3-input truth table? [2 marks]

Question 2

Evaluate (A AND B) OR C when A=1 B=0 C=1. [2 marks]

Question 3

Explain the benefit of an intermediate X column. [2 marks]

Exit Ticket · Answer before checking

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

A three-input truth table has only four rows.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. It is a one-input inverter, usually a triangle with a small output bubble.

Q2. B — Ciphertext

Q3. People may lose control over how identifying information is used or shared.

Q4. Plan clear steps in the right order to solve a problem. Keep needed details, split the problem into parts, then link the steps.

I Do / We Do pages · Guided practice

Q1. Explain why AND is applied last.

You Do pages · Code First

```
A = true
B = false
C = true
if (A AND B) OR C then
    print("Investigate")
endif
```

Expected result: Investigate

You Do pages · Independent questions

Q1. $2^3=8$.

Q2. 1.

Q3. Allows accurate separate evaluation of one gate and helps trace errors in combined expressions.

Exit Ticket · Model answer

False

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

2. Write a complete new answer in GREEN PEN. Show working if needed: